

M.Tech Program

Advanced Industry Integrated Programs

Jointly offered by University and LTIMindTree

Agentic AI-Based Construction Schedule Disruption Analysis and Recovery Planning System

&

An Explainable RAG Framework for Enterprise Policy Change Impact Analysis

Knowledge partner



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Introduction

- Analyzes construction schedule disruptions such as activity delays and resource shortages.
- Identifies affected activities and determines their impact on the overall project completion.
- Generates and optimizes recovery strategies using Agentic AI and OR-Tools.
- Recommends the best recovery plan and provides an understandable explanation using an LLM.

Objectives

- Detect and analyze the impact of construction schedule disruptions on project activities.
- Identify affected activities and dependencies to determine potential project delays.
- Generate and optimize recovery strategies considering time, resources, and scheduling constraints.
- Recommend the most suitable recovery plan with an explainable, evidence-based justification.

Scope

- Covers construction project schedule analysis using activities, durations, dependencies, and resource constraints.
- Handles disruptions such as activity delays and resource shortages and identifies their impact on downstream activities.
- Generates and evaluates alternative recovery strategies using constraint-based optimization.
- Provides an explainable recovery recommendation through an Agentic AI workflow and LLM-based explanation.

Literature Review

- Mathematical programming and constraint programming are used to create schedules that satisfy resource and dependency constraints.
- Genetic Algorithms, Tabu Search, and other heuristics are used to find good schedules faster, especially for large projects.
- These methods update the schedule when disruptions such as delays or resource problems occur.
- Existing methods mainly focus on schedule optimization or rescheduling separately. They have limited support for understanding disruption reports, analyzing their impact, generating recovery options, and explaining the recommended plan.

Summary of Literature

- Existing research provides strong methods for project scheduling and reactive rescheduling, but these are primarily focused on solving the scheduling problem mathematically.
- There is a need for a unified decision-support system that connects disruption identification, impact assessment, recovery planning, and schedule optimization.
- Our system combines LLM-based disruption interpretation with dependency-based impact analysis and constraint-based schedule optimization.
- An agentic and explainable workflow using LangGraph and OR-Tools that transforms a user's disruption report into feasible recovery alternatives and an optimized, explainable recovery plan.

Additional Information

- **Dataset & Scenarios** : PSPLIB J30 (480 instances) provides activities, durations, dependencies, and resource constraints. It is used to generate baseline schedules, which serve as the reference for disruption and recovery analysis.
- **Project Flow** : User Input → Disruption Handling → Impact Analysis → Recovery Strategy
Generation → Schedule Optimization → Results & Explanation
- **Target Users & Domain** : Project Managers, Construction Planners, and Scheduling Engineers in the Construction Project Management domain, supporting schedule monitoring and recovery decision-making.
- **Core Technologies** : React – UI | Python + FastAPI – Backend | LangGraph + LLM – Agentic workflow & disruption understanding | OR-Tools – Schedule optimization | PostgreSQL – Data storage.

Modules to Cover

- **Project & Schedule Management**

- Load project data and baseline schedule
- Manage activities, durations, dependencies and resources

- **Disruption Handler**

- Capture disruption details
- Interpret delay, resource issues and other schedule changes

- **Impact Analyzer**

- Identify the affected activity
- Trace dependent/downstream activities
- Calculate the impact on the project schedule

- **Recovery Strategy Generator**

- Generate possible recovery options
- Consider resource availability, dependencies and project constraints

Modules to Cover

- **Schedule Optimizer**

- Use OR-Tools to create feasible recovery schedules
- Handle resource and dependency constraints

- **Plan Evaluator**

- Compare different recovery plans
- Evaluate makespan, delays and resource usage

- **Recommendation & Explanation**

- Select the best recovery plan
- Use LLM to explain why the plan was selected

- **User Interface & Data Management**

- React dashboard for Project Manager/Admin/Employee
- FastAPI for backend communication
- PostgreSQL for storing projects, disruptions and results